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Electric Vehicle Routing With Recharging Stations: Trade-Offs in Last-Mile Delivery

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ABSTRACT

Last-mile logistics increasingly adopt electric vehicles to address environmental concerns and reduce operational costs. Unlike classical vehicle routing problems, it is essential to consider charging stations in the route planning for electric vehicles. This study aims to investigate the effect of different charging strategies on last-mile delivery optimisation. The adaptive large neighbourhood search (ALNS) algorithm is proposed to solve large-scale problems. The results of the proposed algorithm are compared with the results of the mathematical model in small-scale problems, and the algorithm's performance is proven. The proposed algorithm contributes to electric vehicle route planning by providing effective results in solving large-scale problems. The test problems are solved with three different charging strategies: full charging, partial charging, and partial charging between 20–80% state of charge (SoC). Solutions have been obtained for the objective functions of the minimising total distance, the minimizing total time, and the minimising total energy consumption. The results of the experiments show that the average charging time is the lowest when the total travel time is minimised, the highest values are reached when the total distance is minimised, and more balanced results are provided when the energy consumption is minimised. These findings help logistics companies to determine the most appropriate charging strategy in terms of operational efficiency and cost optimisation.

1 | Introduction

Last-mile delivery, all logistics related to the delivery of shipments to private customer households in urban areas, is a hot topic in cities worldwide. The growth of the urban population and increasing e-commerce activities increase the complexity of last-mile delivery and its impact on the environment and quality of life. Despite its relevance, developing countries struggle to create and implement alternatives to the traditional scope of fossil-based operations. More and more logistics and distribution companies are adopting electric vehicles (EVs) for cargo distribution [1]. With the tightening of environmental protection policies, regulations have been introduced to avoid the distribution of conventional fossil fuel vehicles in urban areas of some cities [2]. However, technical and cost constraints such as long battery charging times, limited driving range, lack of charging infrastructure and high charging costs still hinder the large-scale implementation of EVs [3]. These problems also bring difficulties in vehicle scheduling and route planning. Innovative solutions are needed to address these challenges and promote the widespread adoption of EVs for last-mile delivery.

The development and implementation of innovative solutions for last-mile delivery using EVs is crucial to address the

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environmental impact and logistical challenges associated with traditional fossil fuel-based operations. Various technologies and strategies are proposed to optimize delivery routes, including optimisation algorithms, machine learning techniques, and real-time data analysis, taking into account factors such as traffic conditions, delivery windows and charging station availability [4]. These solutions should focus on improving battery technology to extend driving ranges [5], expanding charging infrastructure in urban areas [6], and implementing efficient vehicle scheduling and route planning algorithms that maximise delivery efficiency while minimising costs and environmental impact [7–9]. By integrating these technologies and strategies, logistics companies can achieve more sustainable and efficient last-mile delivery operations while reducing their carbon footprint. Therefore, developing an optimal scheduling policy, including flexible pricing plans, is essential to not only minimise overall travel time but also lower delivery costs.

The widespread use of EVs in last-mile delivery offers an excellent opportunity to increase the efficiency of logistics operations as well as reduce environmental impacts [10]. In this context, the capacitated electric vehicle routing problem (CEVRP) is a critical problem that enables the planning of optimal routes by considering the capacity and charging requirements of the EVs. As a result of the literature review, it was seen that most of the existing studies focused on full charging or partial charging strategies. In pioneering studies that propose mathematical models, the full charging strategy with fixed charging times and limited battery capacities for electric vehicle routing problem with time windows (EVRPTW) were taken into account [11]. The objective function is to minimise the total distance. A solution with a partial charging strategy for electric vehicle routing problem (EVRP) with time windows was proposed for this objective function in [12]. In our study, route optimization was carried out by considering battery health in partially charging between 20% and 80% state of charge (SoC). Results of test problems are analyzed for three different objective functions by taking into account full charging, partial charging, and 20–80% partial charging strategy so that the SoC does not fall below 20% and does not exceed 80%. In this context, our study differs from the study of [13], which considers the batteries' SoC to determine the performed routes with the minimum time; overcharging and deep discharges are avoided by protecting the battery from degradation.

This study aims to analyse and evaluate the effectiveness of different charging strategies for CEVRP in last-mile delivery. This study considers three charging strategies: full, partial, and partial charging between 20% and 80% SoC. EVs leave the warehouse at full charge during the routing phase. The application will determine the delivery point at which the EV needs to go to the charging station. The EV will be fully charged at the charging station in the full charging scenario. In the partially charging scenario, the EV will be partially charged at the existing charging station, sufficient to meet the remaining delivery demands. In partially charging between 20% and 80% SoC, an EV can charge up to 80% of its battery capacity when it needs charging. The state of charge cannot fall below 20% at any time. It becomes essential to consider charging strategies in route optimisation to minimise travel times and costs. The CEVRP is solved using the adaptive large neighbourhood search (ALNS) algorithm by optimising criteria such as total travel distance, total travel time considering charging time, and total energy consumption. The contributions of this study are as follows:

- Trade-offs between routes obtained for objective functions such as minimising total distance, minimising total time, and minimising total energy are analysed by using different charging strategies within the scope of the CEVRP problem. The results provide practical implications for logistics companies, exhibiting how charging strategies impact total travel time and overall costs in last-mile delivery operations.
- Two different charging strategies (full recharge and partial recharge) are generally used in the literature. In this study, in addition to these charging strategies, a 20–80% battery rule is utilised, which emphasises long-term battery health by limiting charging between 20 and 80%. This rule aims to balance the long-term health of the EV battery and the convenience of a full charge [11].
- By analysing and comparing different charging strategies, the study expands the literature on CEVRP, offering insights into optimizing route planning based on battery constraints and charging requirements.
- Dataset, including the network of the study area, charging station locations and delivery demand locations, is used in the Simulation of Urban Mobility (SUMO) simulation environment via OpenStreetMap (OSM).

The results of this study will provide valuable results for logistics companies to emphasise the effect of charging strategies on various objectives, especially on the total travel time in last-mile delivery operations. These insights and recommendations can guide the development of strategies that not only increase customer satisfaction but also reduce overall travel costs, which decreases delivery costs, ensuring a win-win situation for both logistic companies and customers. The rest of the paper is organised as follows: Section 2 briefly reviews previous research on vehicle routing problems from EVs. Material and methods are given in Section 3. Section 4 discusses the simulation, results, and performance of the proposed system. Section 5 presents the research conclusions and provides an outlook for future work.

2 | Literature Review

2.1 | Electric Vehicle Routing Problem

The vehicle routing problem (VRP) is one of the most important problems encountered by businesses in the distribution of goods and has attracted the attention of many researchers and practitioners. VRP aims to determine the lowest-cost delivery routes from a warehouse to a geographically dispersed set of customers [14]. VRP is divided into various subcategories depending on the target functions and constraints of distribution systems. This diversity gives rise to various problems with different structural and operational characteristics. As environmental concerns increase and sustainable practices become increasingly important, many logistics companies have considered the use of EVs in their fleets to reduce greenhouse gas emissions. Due to environmental concerns, EVs have been employed for last-mile deliveries in several places. The advantages of EVs include lower

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transportation costs, higher energy efficiency, and lower emissions of pollutants. EVs are powered by rechargeable batteries. However, EV routing faces significant difficulties, such as poor battery energy density, inadequate electric charging and battery switch stations, and longer recharging times. These restrictions add to the complexity and difficulty of the EVRP challenge [15, 16].

Mixed integer programming (MIP) has been mostly used to formulate the EVRP. A mixed integer linear programming model for the vehicle routing problem with time windows was formulated in [11]. They proposed the model to minimise the total travel distance. Instead of fully charging, partial charging has been proposed in many studies. The mixed integer linear programming model by considering partial charging strategy for EVRP with time windows was formulated [12]. The literature indicates that metaheuristic algorithms are predominantly used to solve the EVRP. The EVRP problem is an NP-hard problem, and metaheuristic methods are preferred in the literature to obtain solutions in a reasonable time for large-scale real-life problems. A review of the literature reveals that ALNS is the most preferred method among metaheuristics. A variable neighbourhood search (VNS) with the tabu search (TS) metaheuristic was adopted to solve EVRP in [11]. In [17], the EVRP was addressed with partial recharges, considering three recharges —normal, fast, and super-fast—using the ALNS approach. Ant colony optimisation (ACO) metaheuristic method was presented in [18] to optimise total energy consumption in the EV routing problem. A two-stage CEVRP was proposed, incorporating various factors such as battery capacity, load capacity and battery charging cost [19]. They solved the problem using a hybrid metaheuristic algorithm that combines column generation and ALNS. A VNS Algorithm and variable neighbourhood descent algorithm were used in [20]. The ALNS algorithm has also been used to solve the problem. A template-based ALNS algorithm with a two-phase solution scheme was proposed in [21].

Recent research has increasingly focused on integrating EVRP with sustainability goals, particularly within the scope of green logistics, to support decarbonization efforts in transportation [22]. Green logistics location-routing problem with eco-packages (GLLRPE) was addressed in [23]. A multi-depot vehicle routing problem with time windows that includes intelligent recycling prices and transportation resource sharing (MDVRPTW-IRPTRS) was introduced in [24]. The optimal solution of the problem was determined by using a hybrid algorithm that combines a three-dimensional (3D) k-means clustering algorithm with a self-adapting genetic algorithm-particle swarm optimisation (SGA-PSO). A multi-objective mathematical model to optimize the number of vehicles and carbon emissions in [25]. They used the spectral clustering algorithm and improved multi-objective ACO with variable neighbourhood search (SCA-IMOACO-VNS algorithm) to implement the model. A multi-depot vehicle routing problem with time windows that incorporates intelligent recycling prices and transportation resource sharing (MDVRPTW-IRPTR) was addressed in [26]. The problem was formulated as a mixed integer programming (MIP) model to maximise the total recycling profit while minimising the total operating costs. A three-dimensional (3D) k-means clustering algorithm with a self-adapting genetic algorithm particle swarm optimisation (SGA-PSO) algorithm was proposed to solve the problem.

2.2 | Charging Strategy

EVRP studies include various features and constraints such as EV fleet type (homogeneous/heterogeneous), charging strategy (full/partial/full and partial), constraints about recharges per route (single full/single partial/multiple full/multiple partial), public/private stations, charging mode (slow/normal/fast), charging operations at customer locations (yes/no), and waiting times at charging stations (no wait/deterministic waiting times/stochastic waiting times). Due to the limited number of charging stations and the restricted travel range of EVs, they need to recharge their batteries during their route. Two primary recharging strategies are commonly used: full charging and partial charging. In full charging, the EV's battery is always fully charged when visiting a charging station. However, fully charging an EV takes much longer than refuelling a traditional vehicle. The usual technique of recharging a battery to its full condition takes longer and affects the routing of EVs significantly, especially when the consumers set delivery time limits. Partial recharging, which allows the battery to be recharged to the necessary level in less time, is a potential solution to this problem. With partial recharging, an EV can leave the charging station at any charging level, depending on the time spent charging. In pioneering studies that propose mathematical models, the full charging strategy with fixed charging times and limited battery capacities for EVRP with time windows was considered [11]. A solution with a partial charging strategy for EVRP with time windows was proposed in [12]. In [13], the authors were considered the battery's SoC to determine the performed routes with the minimum time; overcharging and deep discharges are avoided by protecting the battery from degradation.

Different charging technologies can be used according to the type of vehicle. Each has different construction costs, operating costs, charging times, and capacities. Some studies have addressed different charging technologies that allow slow or fast charging with several operating costs [27]. In [28], multiple recharging methods, such as partial recharging and battery swapping, were integrated to address the EV routing problem with time window constraints. There are various restrictions, such as vehicle load capacity, battery capacity, charging station capacity, at most a single recharging per route and no waiting at charging stations. In [29], the authors have dealt with the electric vehicle routing problem to find optimal charging schedules and routes. In [20], the EVRP with multiple depots and heterogeneous fleets was proposed. Partial charging was considered in [30]. The problem has included uncertainties such as road conditions or energy consumption in their research. Hybrid electric vehicles have been considered. An efficient coordinated charging method for plug-in hybrid electric vehicles to minimise the grid overloading and voltage issues was investigated in [31]. An electric vehicle routing problem with states of charging stations and an optimisation model was proposed in [32]. A continuous-time model for EVRP with nonlinear charging functions and capacitated stations was proposed in [33]. In [34], authors considered the route optimization problem, the electric fleet size, and the mixed vehicle routing problem with a partial linear recharging strategy (E-FSMFTW-PR). The problem includes a partial linear charging strategy for a fleet of EVs that vary in fixed cost, payload capacity, and battery size. To solve the problem, a hybrid large neighbourhood aearch (LNS) algorithm based on the iterated local search (ILS)

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was proposed. In [25], a bi-objective nonlinear model for EVRP, considering multiple depots, shared charging stations, and time windows, was proposed in order to minimise the number of EVs and total operational costs. They also suggested a hybrid of multi-objective GA and TS algorithms for solving this problem. The results reveal that the algorithms can reduce operational costs.

3 | Material and Methods

3.1 | Capacitated Electric Vehicle Routing Problem

EVRPs bring difficulties in vehicle route planning and task scheduling because of technical and cost constraints such as long battery charging times, limited driving range, lack of charging infrastructure, and high charging costs. It is essential to reduce costs in last-mile-delivery-specific routing services. This study handles the CEVRP problem, which is important for third-party logistics firms in last-mile delivery. Although there is extensive research on the CVRP for conventional fuel vehicles, studies on EVs are limited. Each vehicle route may include visits to charging stations. Assumptions for CEVRP are as follows:

- EVs are homogeneous.
- The payload capacity of the vehicle cannot be exceeded.
- Each EV route starts from a depot to serve customers and ends at the same depot.
- All EVs in the fleet leave the depot at full charge.
- The time is known for a full charge.
- The locations of the charging stations and the travel distance from any node to any charging station are known.
- Each customer node will be serviced by exactly one EV.
- Multiple packages can be delivered to a customer.
- EVs can visit a charging station between any two customers.
- Each charging station can be visited by more than one EV.

3.2 | Mathematical Model

Optimal solutions are obtained using the mathematical model in [12] for the CEVRP for the full charging strategy. The model has been revised and is being used for different charging strategies.

Sets

0, N+1 *Depot the vehicle leaves from and returns to*

F' *Set of visits to recharging stations, dummy vertices of set of recharging stations F*

F'_0 *Set of recharging stations and depot 0*

V *Set of customers.*

V' *Set of customers and recharging stations*

V'_0 *Set of customers, recharging stations and depot 0*

V_0 *Set of customers and depot 0*

$V'_{0,N+1}$ *Set of customers, recharging stations and Depot 0 and Depot N+1*

V'_{N+1} *Set of customers, recharging stations and Depot N+1*

Parameters

d_{ij} *Travel distance between vertices i and j*

t_{ij} *Travel time between vertices i and j*

C *Vehicle capacity*

g *Recharging rate*

h *Charge consumption rate*

Qc *Battery capacity*

q_i *Demand of customer i*

e_i *Earliest start time of service at vertex i (start of shift)*

l_i *Latest start time of service at vertex i*

s_i *Service time of customer i*

Decision Variables

$$x_{ij} = \begin{cases} 1, & if\ arc\ (i,j)\ is\ traveled \\ 0, & in\ other\ cases \end{cases}$$

p_i *Time of arrival at vertex i*

u_i *Remaining cargo capacity on arrival at vertex i*

y_i *Remaining battery capacity on arrival at vertex i*

Mathematical Model:

$$min \sum_{i\varepsilon V'_0, j\varepsilon V'_{N+1}, i\neq j} d_{ij}x_{ij} \tag{1}$$

s.t.

$$\sum_{j\varepsilon V'_{N+1}, i\neq j} x_{ij} = 1\ \forall i\varepsilon\ V \tag{2}$$

$$\sum_{j\varepsilon V'_{N+1}, i\neq j} x_{ij} \leq 1\ \forall i\ \varepsilon\ F' \tag{3}$$

$$\sum_{i\varepsilon V'_{N+1}, i\neq j} x_{ji} - \sum_{i\varepsilon V'_0, i\neq j} x_{ij} = 0 \quad \forall j\ \varepsilon\ V' \tag{4}$$

$$P_i + \left(t_{ij} + s_i\right) x_{ij} - l_0\left(1 - x_{ij}\right) \leq P_j \quad \forall i\ \varepsilon\ V_0, \forall j\ \varepsilon\ V'_{N+1}, i \neq j \tag{5}$$

$$P_i + t_{ij}x_{ij} + g\left((Qc) - y_i\right) - \left(l_0 + g\left(Qc\right)\right)\left(1 - x_{ij}\right) \leq P_j$$
$$\forall i\ \varepsilon\ F', \forall j\ \varepsilon\ V'_{N+1}, i \neq j \tag{6}$$

$$e_j \leq P_j \leq l_j\ \forall j\ \varepsilon\ V'_{0,N+1} \tag{7}$$

$$0 \leq u_j \leq u_i - q_i x_{ij} + C\left(1 - x_{ij}\right) \quad \forall i\ \varepsilon\ V'_0, \forall j\ \varepsilon\ V'_{N+1}, i \neq j \tag{8}$$

$$0 \leq u_0 \leq C \tag{9}$$

$$0 \leq y_j \leq y_i - (d_{ij}h)x_{ij} + Qc\left(1 - x_{ij}\right) \quad \forall i\ \varepsilon\ V, \forall j\ \varepsilon\ V'_{N+1}, i \neq j \tag{10}$$

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$$0 \leq y_j \leq Qc - (d_{ij}h)x_{ij} \quad \forall i \in F'_0, \forall j \in V'_{N+1}, i \neq j \tag{11}$$

$$X_{ij} \in \{0,1\} \ \forall i \in V'_0, \forall j \in V'_{N+1} \ i \neq j \tag{12}$$

The total travelled distance is minimised with the objective function (1). Constraint (2) ensures that service is provided to every customer. Constraint (3) limits each charging station to be used only once along the route. Constraint (4) is the sub-tour blocking constraint. Constraint (5,6) guarantees time feasibility. Constraint (7) ensures distribution during working hours. With constraint (8), when the EV leaves a node, it ensures that its load is equal to or greater than the demand of the next node. Constraint (9) ensures that the EV load does not exceed its capacity when leaving the warehouse. Constraint (10, 11) calculates the remaining charge of EVs when they arrive at a node. Constraint (12) shows the binary decision variable indicating if $arc\ (i, j)$ is travelled. The decision variable will take an integer value of 0 or 1.

3.3 | Proposed Adaptive Large Neighbourhood Search Algorithm

Metaheuristic algorithms are frequently used in complex optimisation problems, such as CEVRP, where multiple constraints must be considered. ALNS is one of the metaheuristic algorithms used to solve these problems [35]. ALNS can quickly search the solution space with its adaptive and dynamic operator structure. The proposed ALNS algorithm has a flexible and dynamic structure that can evaluate different charging strategies (full charge, partial charge, 20%-80% SoC) for different objective functions for large-scale EVRP problems. The effects of charging strategies such as 20–80% SoC to protect battery health on operational results have rarely been addressed in the literature. Most studies have focused only on full charge or fixed charge times. This study offers a significant innovation by analysing the effects of charging strategies on route planning, which are generally ignored in the literature. In the literature, the focus has generally been on optimising a single objective function. However, the balance between total energy consumption, travel time, and distance has not been sufficiently investigated. The ALNS algorithm proposed in this study saves energy and time in large-scale EVRP problems and provides more effective results than small-scale mathematical models. Thanks to these innovations, our algorithm provides a more effective balance between total distance, total energy consumption and total time optimisation by considering different charging strategies. It also provides faster and more effective solutions for large-scale problems.

The ALNS algorithm starts with an initial solution and generates neighbouring solutions at each step using different destroy and repair operators. Whether new solutions are accepted or rejected is determined by evaluating them against an acceptance criterion. The flowchart of the algorithm is shown in Figure 1.

The ALNS algorithm consists of specific steps and components. First, an initial solution is determined, which is expanded by applying certain destruction and repair operators. A new solution is obtained in each iteration. The new solution is evaluated against the previous solution. S is the current solution, S' is the new solution and S^{best} is the best solution. This study proposes a new ALNS algorithm by integrating Local Search (LS) and Simulated Annealing (SA) (Algorithm 1).

In Algorithm 1, the cost function $f(S)$ is calculated for each objective function as follows:

Assume that the solution S *is*

$$S = cs5 \rightarrow 35 \rightarrow cs7 \rightarrow 115 \rightarrow 2 \rightarrow 32 \rightarrow cs5$$

If the cost function is total distance, then $f(S)$ is calculated as follows:

$$\text{cs5} \rightarrow 35 \rightarrow \text{cs7} \rightarrow 115 \rightarrow 2 \rightarrow 32 \rightarrow \text{cs5}$$

$$d_{cs5,35} \quad d_{35,cs7} \quad d_{cs7,115} \quad d_{115,2} \quad d_{2,32} \quad d_{32,cs5}$$

$$f(S) = \sum_{(i,j)\in E'} d_{i,j} = 3158.68$$

where $d_{i,j}$ is travel distance from node i to node j, and E' is the edge list on the the S.

If the cost function is total time, then $f(S)$ is calculated as follows:

$$\text{cs5} \rightarrow 35 \rightarrow \text{cs7} \rightarrow 115 \rightarrow 2 \rightarrow 32 \rightarrow \text{cs5}$$

$$t_{cs5,35} \quad t_{S_{35}} \quad t_{35,cs7} \quad t_{C_{cs7}} \quad t_{cs7,11} \quad t_{S_{115}} \quad t_{115,2} \quad t_{S_2} \quad t_{2,32} \quad t_{S_{32}} \quad t_{32,cs5}$$

$$f(S) = \sum_{(i,j)\in E'} t_{i,j} + \sum_{i\in C'} t_{S_i} + \sum_{k\in CS'} t_{C_k} = 2653.07$$

where $t_{i,j}$ is travel time from node i to node j, t_{S_i} is service time of node i, t_{C_k} is recharging time at charging station k, C' is the customer list, and CS' is the charging station list on the S.

If the objective function is total energy consumption, then $f(S)$ is calculated as follows:

$$\text{cs5} \rightarrow 35 \rightarrow \text{cs7} \rightarrow 115 \rightarrow 2 \rightarrow 32 \rightarrow \text{cs5}$$

$$en_{cs5,35} \quad en_{35,cs} \quad en_{cs7,115} \quad en_{115,2} \quad en_{2,32} \quad en_{32,cs5}$$

$$f(S) = \sum_{i,j} en_{i,j} = 338.68$$

where $en_{i,j}$ is energy consumption from node i to node j.

3.3.1 | Initial Solution Generation

Various methods are used in the literature to generate the initial solution. Creating completely random solutions makes it difficult to converge to optimal solutions. Therefore, in the initial solution, the route is designed by selecting customers closest to each other, using the nearest neighbour heuristic. Each route is created while considering the vehicle capacity constraint. The pseudocode for the initial solution is given in Algorithm 2.

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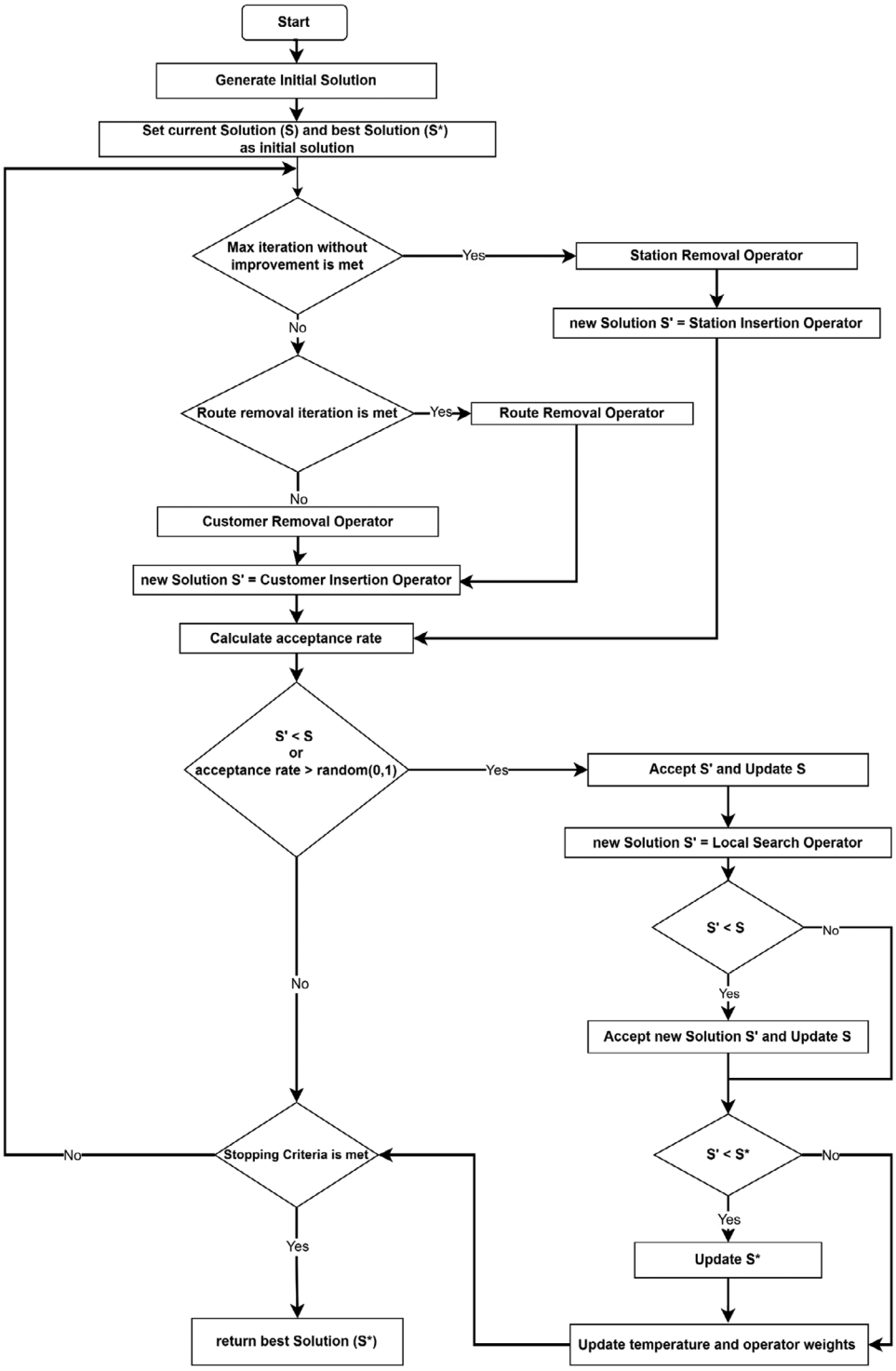

FIGURE 1 | Proposed ALNS algorithm.

The initial solution is destroyed and repaired again in each iteration by the operators until the termination criterion is met. This process ensures that new neighbouring solutions are generated at each time. A new neighbouring solution is obtained and evaluated against the acceptance criteria at each stage. ALNS can handle customers and charging stations to obtain new solutions. It can add charging stations to the route where necessary or remove them from routes where they are not needed. Similarly, customers or routes can be removed and rearranged to explore better solutions. The use of different operators and the effectiveness of these operators are critical for obtaining higher quality solutions.

3.3.2 | Customer Removal and Insertion Operators

Customers removed from the route are kept in a list of unserved customers. After the destruction process, the repair operators

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ALGORITHM 1 | The proposed ALNS with LS operators

input: Initial Solution S_{init}, Ω^-, Ω^+, Ω^{Ls}, N, K, Z, T_0, α

Initialize i, j, w^+, w^-, w^{Ls}, p^-, p^+, p^{Ls}

$S^* \leftarrow S_{init}$, S' $\leftarrow S_{init}$, S $\leftarrow S_{init}$, T $\leftarrow T_0$, i $\leftarrow$ 1, j $\leftarrow$ 0

repeat

  if i = = 0 (mod K) **then**

  Select route *removal* and customer *insertion* operator Ω^-, Ω^+ using Roulette Wheel Selection

  S' $\leftarrow$ *insert(remove(S))*

  else

    if j = = 0 (mod N) **then**

  Select station *removal* and *insertion* operator Ω^-, Ω^+ using Roulette Wheel Selection

  S' $\leftarrow$ *insert(remove(S))*

    else

  Select customer *removal* and *insertion* operator Ω^-, Ω^+ using Roulette Wheel Selection

  S' $\leftarrow$ *insert(remove(S))*

    end if

  end if

  acceptance rate $\leftarrow e^{(-(f(S')-f(S))/T)}$

  if *f(S') < f(S)* or acceptance rate > *random*(0,1) **then**

  S $\leftarrow$ S'

  Select *local search* operator Ω^{Ls} using Roulette Wheel Selection

  new S' $\leftarrow$ *local search(S)*

    if *f(new S') < f(S)* **then**

  S $\leftarrow$ new S'

    end if

    if *f(S) < f(S*)* **then**

  S* $\leftarrow$ S

    end if

  j $\leftarrow$ 0

  else

  j $\leftarrow$ j + 1

  end if

  if i = = 0 (mod Z) **then**

  Update w^+, w^-, w^{Ls}, p^-, p^+, p^{Ls}

  end if

  T $\leftarrow$ T * α

  i $\leftarrow$ i + 1

until stopping criteria is met;

return S*

ALGORITHM 2 | Initial solution

input: Depot d, List of unserved customers $C_{unserved}$, Vehicle capacity Q

Initialize: initial solution $S_{init} \leftarrow \emptyset$, $C_{served} \leftarrow \emptyset$

while $|C_{unserved}| > 0$ **do**

route $\leftarrow$ [d]

  $c_{min} \leftarrow \min\{c \in C_{unserved}\}$ distance(d, c)

  route $\leftarrow$ route $\cup \{c_{min}\}$

  $C_{served} \leftarrow C_{served} \cup \{c_{min}\}$

  $C_{unserved} \leftarrow C_{unserved} \setminus \{c_{min}\}$

while (load(r) $\leq$ Q) and ($|C_{unserved}| > 0$) **do**

    $c_{next} \leftarrow \min\{c \in C_{unserved}\}$ *distance*((last(*route*), c)

    route $\leftarrow$ route $\cup \{c_{next}\}$

if load(r) > Q **then**

      route $\leftarrow$ route $\setminus \{c_{next}\}$

break

else

      $C_{served} \leftarrow C_{served} \cup \{c_{next}\}$

      $C_{unserved} \leftarrow C_{unserved} \setminus \{c_{next}\}$

end if

end while

  route $\leftarrow$ route $\cup$ {d}

  $S_{init} \leftarrow S_{init} \cup$ {route}

end while

return S_{init}

TABLE 1 | Customer removal operators.

Operator	Description
Related removal	P customers close to a selected starting customer are removed from the route. This method targets clustered customers.
Random removal	A random number of p customers are removed from the route. This operator provides randomness in the solution space, allowing for wider scanning.
Worst removal	P customers who have a high cost of being on the route or who will cause the greatest cost gain when removed from the route are removed from the route.

consider these unserved customers to be reinserted into a route. The customer removal and insertion operators are given in Tables 1 and 2, respectively.

3.3.3 | Route Removal Operators

Route removal operators remove all customers from a selected route at longer iteration intervals and add them to the list of

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TABLE 2 | Customer insertion operators.

Operator	Description
Greedy insertion	Unserved customers are added to the places that provide the best cost on the routes with a greedy approach.
Regret-2 insertion	Using the greedy approach can create high costs for subsequent unserved customers, thus moving the solution away from optimality. To prevent this, the Regret-2 method is used. The regret value, which is the difference between the best placement and the second-best placement, is calculated, and the customer with the higher value is selected.

TABLE 3 | Route removal operators.

Operator	Description
Random removal	A random route is selected from the solution, and all customers on that route are removed from the route and added to the list of unserved customers.
Worst removal	The routes in the entire solution are examined for their total cost, and the route with the highest cost is destroyed. The customers on that route are added to the list of unserved customers.

unserved customers. Performing route removal operators too frequently can negatively affect the solution time. The route removal operators are given in Table 3.

After the route is removed, unserved customers occur. Customer insertion operators are used to add these customers back to the routes. Thus, different customer combinations are scanned in the solution space.

3.3.4 | Station Removal and Insertion Operators

EVs have a limited driving range by nature. Therefore, stopping at a charging station on the route may be necessary. Questions such as which station to stop at and how long to stop are critical. In the ALNS algorithm, this is provided by station insertion/removal operators. Station removal operators are given in Table 4.

After using the station removal operator, the station insertion operator is used. Table 5 shows the station insertion operators.

3.3.5 | Local Search Operators

In this study, local search (LS) methods are used to explore the solution space within the scope of the study better. These methods aim to improve the solution by searching the neighbourhoods of the current solution in a different way. LS methods are divided into two categories: intra-route and inter-route. Intra-Route focuses on improving solutions by changing the positions of customers within a single route. Inter-Route aims to improve solutions by changing the positions of customers between different routes. The LS used in the study are implemented as operators. These operators are listed in Table 6.

TABLE 4 | Station removal operators.

Operator	Description
Worst charge usage removal	Removes p of the lowest utilisation charging stations from the route. Station utilization is calculated based on the charging time on the route.
Worst station removal	Stations on the route are examined according to their distance to customers, and the station with the highest distance cost is removed.
Random station removal	*P* randomly selected stations are removed from the routes.

TABLE 5 | Station insertion operators.

Operator	Description
Greedy station insertion	The station that provides the least distance cost is inserted between the customer point where the charging capacity of the electric vehicle is negative and the previous customer. If no station can be inserted to this point, the insertion process is tried at the previous customer point.
Best station insertion	For the customer point and all points before it where the charging capacity of the electric vehicle is negative, it examines the station insertion alternatives and makes the insertion with the best cost.
Random station insertion	The nearest station is inserted after a randomly selected customer on the route.

The algorithm selects from a specific group of operators in each iteration. Each operator has a weight value. Initially, Ω_+ defines the repair operator set, Ω_- defines the destroy operator set, and the weights are calculated with the formulas $\frac{1}{\Omega_+}$ and $\frac{1}{\Omega_-}$. The algorithm selects the operator from the operator pool using the roulette wheel selection method. This method is given in Equation 13.

$$\mathrm{P}(\mathrm{i}) = \frac{w(i)}{\sum_{i' \in \Omega} w(i')} \tag{13}$$

The ratio of the weight value $w(i)$ of an operator to the total weights in the operator set gives the probability of selecting that operator. This probability value changes depending on the operator's weight. The weights of the operators change according to their performance status. In this respect, it has a dynamic and

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TABLE 6 | Local search operators.

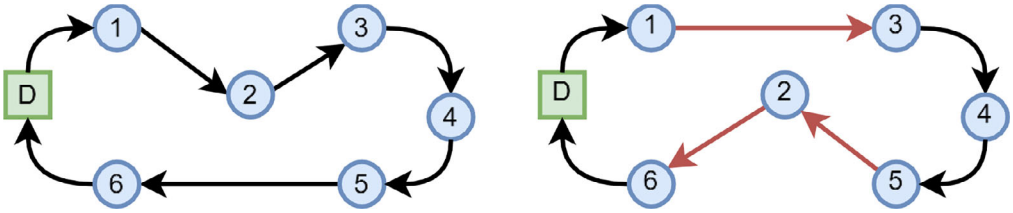
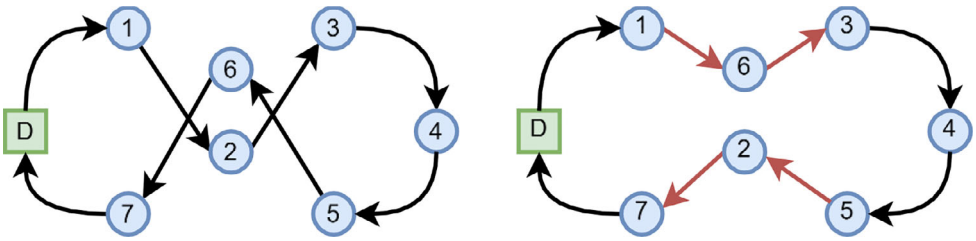
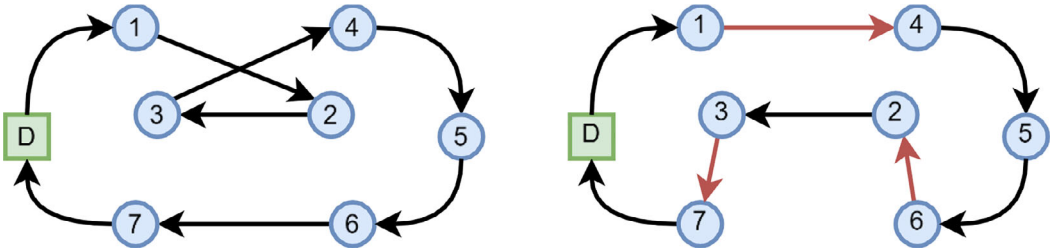
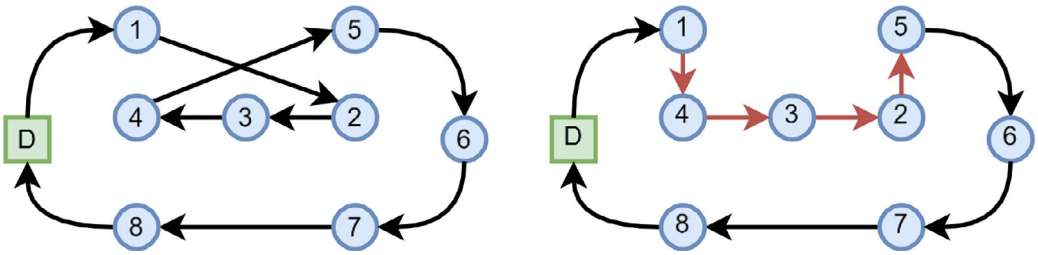
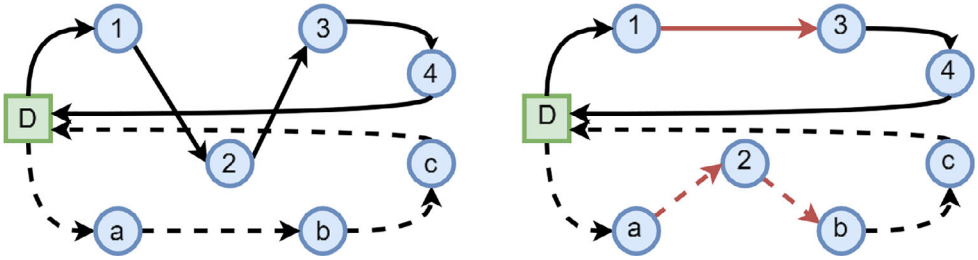
Operator	Description
IntraRelocate	A customer on a route is moved to another position on the route. This process is intended to reduce the cost of the route.
	
IntraExchange	Two customers selected from the route swap positions within the route.
	
IntraOrOpt	Several consecutive customer segments selected from the route are moved to another position within the route.
	
IntraTwoOpt	Two selected customers in the route are arranged in reverse order in the route.
	
InterRelocate	The process of transporting a selected customer between two different routes.
	

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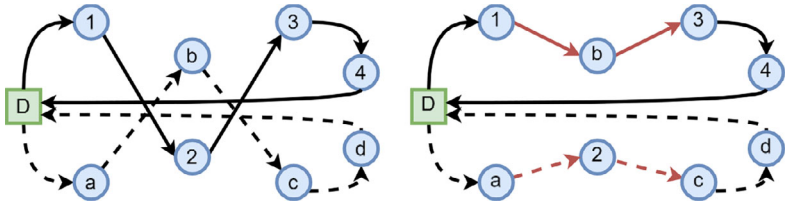
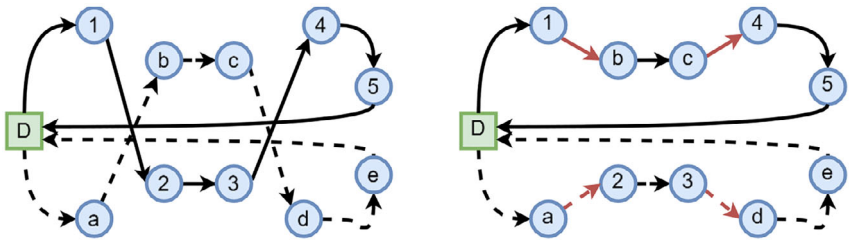
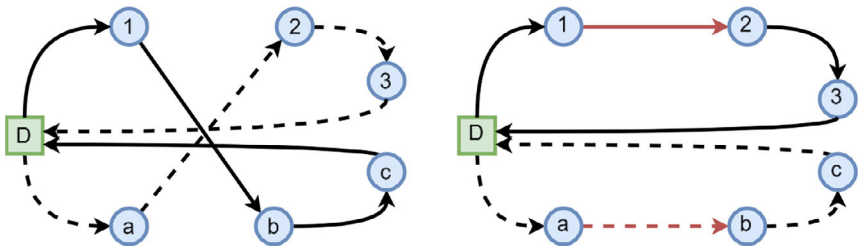
TABLE 6 | (Continued)

Operator	Description
InterExchange	The process of swapping places between two customers selected from two different routes.
	
InterCrossExchange	The process of mutually exchanging the customer segment selected between two different routes.
	
Inter2Opt*	The process of swapping the places of selected customers on two different routes diagonally with each other on the following routes.
	

flexible structure. Previous performances are effective in selecting the operator. For this process, operators earn scores with certain iterations. This score calculation is given in Equation 14.

$$\pi_i = \begin{cases} \pi_1, & \text{if } S' < S^{best} \\ \pi_2, & \text{if } S^{best} < S' < S \\ \pi_3, & \text{if } S' > S \\ 0, & \text{if } S = \emptyset \end{cases} \tag{14}$$

If the used operator produces a solution better than the best solution, π_1, if it improves the current solution but is worse than the best solution, π_2, if it is an acceptable solution, π_3, if a solution cannot be obtained or it is an infeasible solution, the score is 0. These scores will change the operator weights and therefore the probability of being selected. Equation 15 is used when updating the weights with the obtained scores.

$$\mathrm{w(h)} = \begin{cases} (1-r)\cdot w(i) + r\cdot \pi_i/Q_i & \text{if } Q_i > 0 \\ (1-r)\cdot w(i) & \text{if } Q_i = 0 \end{cases} \tag{15}$$

This process is called adaptive weight adjustment. When determining the new weight *w(h)*, the Q_i value indicates how many times the ith operator is used, and π_i indicates the score of the ith operator. The *r* parameter decides whether the score or the previous weight value *w(i)* will be dominant in updating the weight. This value is a probability value between (0, 1) and is important in terms of parameter optimisation. If a high value is given, the obtained score will be dominant over the new weight.

3.3.6 | Acceptance Criteria

After the operators are applied, the new neighbouring solutions obtained are evaluated with the acceptance criterion. The acceptance criterion decides whether the new solution will be accepted or not. In this study, new solutions are evaluated using the Metropolis Criteria. If the newly obtained solution is better than the existing one, the solution is accepted. However, in the case of a bad solution, the algorithm can accept the solution with a certain probability approach. This process is also given in Equation 16 and Equation 17.

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ALGORITHM 3 | SA acceptance criteria.

input: Initial solution S_{init}, cost function f(), initial temperature T_0, cooling rate α

S* ← S_{init}, S' ← S_{init}, S ← S_{init}, T ← T_0

repeat

  S' ← *generate new solution(S)*

  acceptance rate ← $e^{(-(f(S')-f(S))/T)}$

  if *f(S') < f(S)* **or** acceptance rate > *random*(0,1) **then**

    S ← S'

    if *f(S') < f(S*)* **then**

      S* ← S

    end if

  end if

  T ← T * α

until stopping criteria is met

return S*

$$\Delta E = f(S') - f(S) \tag{16}$$

$$P = e^{(-\Delta E/T)} \tag{17}$$

This structure, also known as the simulated annealing (SA) acceptance criterion, converts the difference between the new S' solution and the existing S solution into a probability P with the function $e^{(-\Delta E/T)}$ depending on the temperature value T. This calculated value is compared with a randomly generated probability value $(0, 1)$. If the solution is not feasible but the p value is greater than the random probability value, the solution is accepted. Thus, even if the solution is bad, the new solution is given a chance. The process allows a wider scanning of the solution space. If the T value is high, the P probability converges to 1. This provides a greater chance for bad solutions at high temperatures. As the temperature decreases, the P probability converges to 0, and each solution is expected to be better than the existing one. In this case, bad solutions are not given a chance. The initially given T temperature is reduced by α in each iteration. This value is called the cooling rate. Equation 18 is used to update the temperature.

$$\mathrm{T}(\mathrm{t}+1) = \alpha \cdot \mathrm{T}(\mathrm{t}) \tag{18}$$

This process ensures that the temperature is reduced by α at each iteration. The pseudocode for the acceptance criterion is given in Algorithm 3.

3.4 | Verification of ALNS Algorithm

To test ALNS algorithm performance, ESOGU problem sets with 5, 10, 20, 40 and 60 customers, categoriSed as Vlustered (C), random (R), and randomly clustered (RC), are used [36]. The proposed ALNS algorithm achieved optimal results in small-scale problems. In large-sized problems, it gave better results in a very short time than the solution given by the mathematical model after 3 h (Table 7). During obtaining a solution with ALNS, stopping by more than one charging station during the route and stopping by a charging station after leaving the depot are prevented.

When the results given in Table 7 are examined, the ALNS achieves optimal results provided by the mathematical model for small-scale problems (C5, R5, RC5, C10, R10, RC10). In these test problems, both the mathematical model and the ALNS algorithm found almost the same distance, and when the solution times are compared, it is seen that the solution times of the ALNS algorithm are much shorter for the problem sets C10, R10 and RC10. In large-scale problems, the mathematical model could not provide optimal solutions within a reasonable time. The results obtained from the CPLEX solver after 3 h are given in Table 7. The ALNS algorithm provided a great advantage in terms of solution time and distance for large-scale problems. For example, in theC60, R60, and RC60 problems, the ALNS algorithm gives better results than the mathematical model with a difference of 7.31, 5.04 and 6.53%, respectively. In addition, the proposed ALNS algorithm is much faster than the mathematical model in terms of solution time. These results prove that the ALNS algorithm reaches a good solution in a reasonable time compared to the mathematical model. It is understood that the proposed ALNS algorithm is a very effective method in terms of solution time, especially for large-scale problems.

Additionally, the superiority of the proposed ALNS algorithm has been demonstrated by comparing it with traditional metaheuristic algorithms (TS, SA). The comparison results are presented in Table 8.

To provide a fair comparison of the three metaheuristic algorithms, the best solutions obtained by each algorithm and the corresponding computational times are presented in Table 8. The time limits for obtaining these solutions were set as follows: 5 s for problems with five and ten customers, 10 s for problems with 20 customers, 20 s for problems with 40 customers, and 30 s for problems with 60 customers. As shown in Table 8, the ALNS algorithm consistently outperforms the TS and SA algorithms in terms of solution quality. For example, while a 3956.43 value was obtained with ALNS for the c05 problem, it was calculated as 3989.19 with TS and SA algorithms. Moreover, as the problem size increases, the proposed hybrid ALNS algorithm demonstrates superior performance not only in terms of solution quality but also in computational time. For example, while a 26158.38 distance value was calculated with ALNS for the r60 problem, 28536.33 and 28720.05 values were obtained with TS and SA, respectively. It is seen that ALNS is more effective in optimising routes compared to SA and TS algorithms. ALNS showed better performance, especially in large-scale problems. Specifically, the results indicate that ALNS provides better solutions within the given time constraints compared to TS and SA, highlighting its effectiveness in solving large-scale optimisation problems.

4 | Experimental Results

The test environment consists of a complete directed graph $G(V, E)$, where E is a set of edges and V is a set of nodes. The edges represent the two-way movable roads/streets. The nodes

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TABLE 7 | Comparison results of the mathematical model and the proposed ALNS algorithm for minimum total distance objective function.

	Mathematical model results (Optimal solutions)		ALNS results		Difference
Test problem	**Distance**	**CPLEX Time**	**Distance**	**Solution time**	**Distance (%)**
C5	3956	0.50	3956.43	2.84	0.00
R5	4977	0.53	4977.08	2.39	0.00
RC5	4944	0.41	4944.1	2.43	0.00
C10	5646	124.33	5645.72	3.24	0.00
R10	4889	389.74	4889.18	3.57	0.00
RC10	5843	345.63	5842.88	3.70	0.00
C20	*9332	11160.69	9124.19	5.64	2.28
R20	*9975	11134.66	9975.22	6.21	0.00
RC20	*9226	11113.25	9161.47	6.15	0.70
C40	*15063	10800.05	16101.75	13.79	−6.45
R40	*18276	10800.02	18274.86	14.72	−1.86
RC40	*17935	10800.02	17051.2	11.34	−2.72
C60	*24817	10800.02	26774.63	17.31	−7.31
R60	*24837	10800.03	26158.38	28.64	−5.04
RC60	*26011	10800.02	27910.51	27.58	−6.53

*It is not the optimal solution. In large-scale problems, the mathematical model could not provide solutions in a reasonable time. The results obtained from the CPLEX solver after 3 h are given.

TABLE 8 | Comparison of the ALNS with traditional metaheuristic algorithms.

	ALNS results		TS results		SA results	
Test problem	**Distance**	**Solution time**	**Distance**	**Solution time**	**Distance**	**Solution time**
c05	**3956.43**	2.84	3989.19	2.59	3989.19	3.29
r05	**4977.08**	2.39	5231.22	2.71	5231.22	2.91
rc05	**4944.1**	2.43	6194.86	2.74	5341.66	3.16
c10	**5645.72**	3.24	6265.07	3.39	6265.07	3.77
r10	**4889.18**	3.57	4922.85	3.92	**4889.18**	4.03
rc10	**5842.88**	3.70	6319.25	4.09	6319.25	4.11
c20	**9124.19**	5.64	9627.14	9.76	10344.52	9.50
r20	**9975.22**	6.21	10459.45	10.07	10408.45	9.58
rc20	**9161.47**	6.15	10881.68	11.31	9635.06	10.41
c40	**16101.75**	13.79	17702.81	18.36	18609.68	17.43
r40	**18274.86**	14.72	21532.91	20.54	20414.57	19.32
rc40	**17051.2**	11.34	19887.18	20.46	19695.46	19.11
c60	**26774.63**	17.31	27599.08	28.27	29473.66	29.17
r60	**26158.38**	28.64	28556.33	30.17	28720.05	30.93
rc60	**27910.51**	27.58	29599.95	30.20	30175.54	31.75

represent the entry and exit of the depot, road intersections, charging stations, and delivery points (the customer's doorsteps). This study implements Eskişehir Osmangazi University Meşelik Campus in the Simulation of Urban Mobility (SUMO) simulation environment via OpenStreetMap (OSM). After the environment is taken from OSM, a graph is created that includes customer demand points, charging stations, and a depot. The representation of the test environment is given in Figure 2.

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TABLE 9 | The problem set with 10 customers (C10).

Point ID	Location	Earliest start time to service	Latest start time to service	Service time	Demand
22A	39.750725, 30.489306	1476	1540	240	95
13	39.750779, 30.491320	1362	1437	120	38
14	39.750497, 30.491334	1662	1668	120	57
45	39.750728, 30.486316	996	1064	180	38
24	39.750821, 30.488751	366	427	120	38
31	39.752950, 30.483109	1372	1448	120	57
34	39.752970, 30.487338	34	78	240	38
26	39.753624, 30.489987	474	531	120	76
113	39.753067, 30.493342	1175	1248	100	38
119	39.752968, 30.486318	858	906	120	95

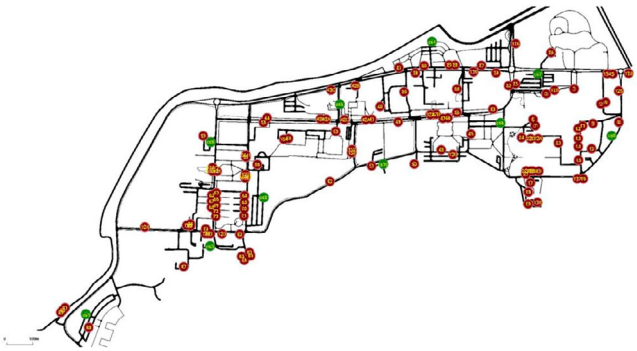

FIGURE 2 | The graph representation of the Eskişehir Osmangazi University.

According to the directed graph $G(V, E)$ in Figure 2, V represents the nodes, customer demand points (1–115), charging stations (cs1-cs10), and depots (cs5), and E represents the edges. Problem sets containing C, R, and RC 5, 10, 20, 40 and 60 customers were defined with the nodes selected from the created graph.

4.1 | Trade-Offs for Objective Functions

The CEVRP with ten customers (Table 9) is solved using three objective functions with the proposed ALNS algorithm to demonstrate the trade-offs between these functions. The results for a test problem are presented in Figure 3 to illustrate how trade-offs are determined on a sample test problem (C10).

The routes with minimum total distance, minimum total energy and minimum total time using proposed ALNS for the C10 problem set are demonstrated in Figure 3.

In the solution where the total distance is optimised, on the first route, the EV stops at the CS8 charging station and remains charging for 280.69 sec in the C10 test problem. For the objective function where the total time is minimised, on the first route, the EV stops at the CS6 charging station and remains charging for 106.41 s.

Table 10 is provided to evaluate the overall results for three different objective functions (minimum total distance, minimum total energy and minimum total time) under the full charging strategy. The optimised results for each objective are analysed in terms of total travel distance, total energy consumption and total travel time.

As seen in Table 10, the total distance when optimizing the total distance objective function is 184114.55 metres, whereas minimising the total energy objective function results in a total distance of 190067.23 metres. When the objective functions of minimising total distance and minimising total energy are compared, it is seen that approximately 5% energy is saved in all solutions for the full charging strategy. When we compare the solutions related to minimise total time with total energy consumption, energy savings exceed 10%. The energy savings increase when considering a partial charging strategy. Each objective function optimses a specific target while introducing trade-offs in the other two factors. Trade-offs are important for decision-makers in logistics companies. To get an overview of the trade-offs, all 15 test problems are solved for three different objective functions and three different charging strategies. Table 11 summarises the trade-offs based on objective functions and charging strategies.

Table 11 provides important information for logistics companies and decision-makers by showing the trade-offs between different objective functions and charging strategies. For instance, when the objective is to achieve minimum total energy consumption, the FC strategy may be the most appropriate choice; however, this requires trade-offs in terms of total distance and total time. Depending on which objective is more important, one should consider trade-offs when selecting a charging strategy. This table helps decision-makers choose the strategy that best suits their goals.

4.2 | Trade-Offs for Recharging Strategy

The results obtained with the proposed ALNS algorithm for the C60 problem set (sixty customers) are shared to demonstrate the frequency of stops at charging stations based on recharging

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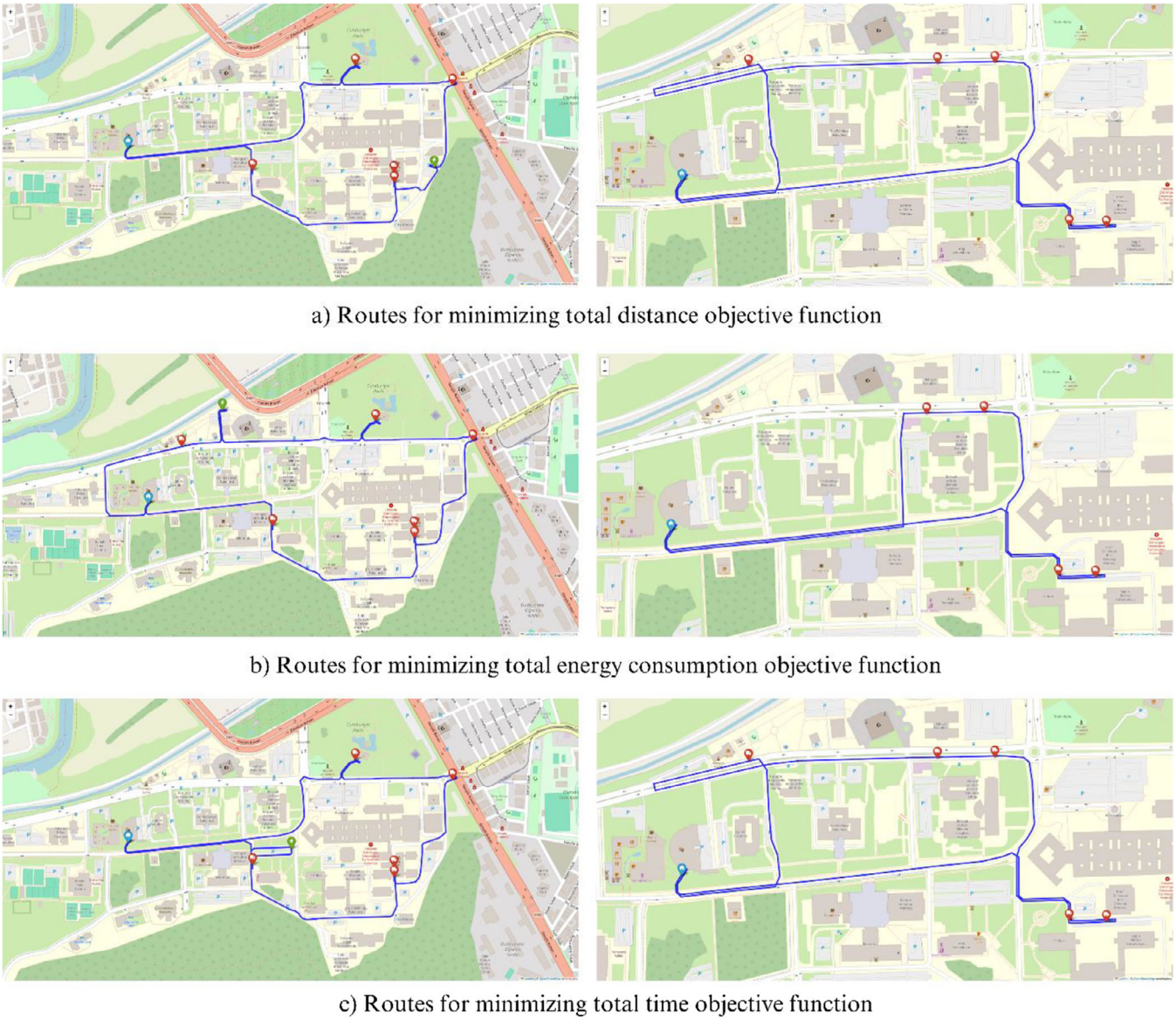

FIGURE 3 | The representation of the routes according to the objective functions.

TABLE 10 | The overall results for three different objective functions.

	Solutions for objective functions		
	Min. total distance	**Min. total energy**	**Min. total time**
Total distance (m)	**184114.55**	190067.23	189871.17
Total energy (kWs)	19571.29	**18648.63**	20578.01
Total time (s)	81505.59	83412.35	**775660.89**

TABLE 11 | Trade-offs related to objective functions and recharging strategies for all problems in ESOGU dataset.

	Min. total distance objective			Min. total time objective			Min. total energy objective		
	FC	**PC**	**20-80**	**FC**	**PC**	**20-80**	**FC**	**PC**	**20-80**
Total distance	184114.55	184726.01	190616.89	189871.17	196909.35	198726.76	190067.23	189368.0	193501.03
Total time	81505.59	79706.98	83484.48	77566.08	77888.74	84091.82	83412.35	80937.83	84835.27
Total energy	19571.29	19859.57	20442.45	20578.01	20835.10	21421.67	18648.63	18760.84	19370.47

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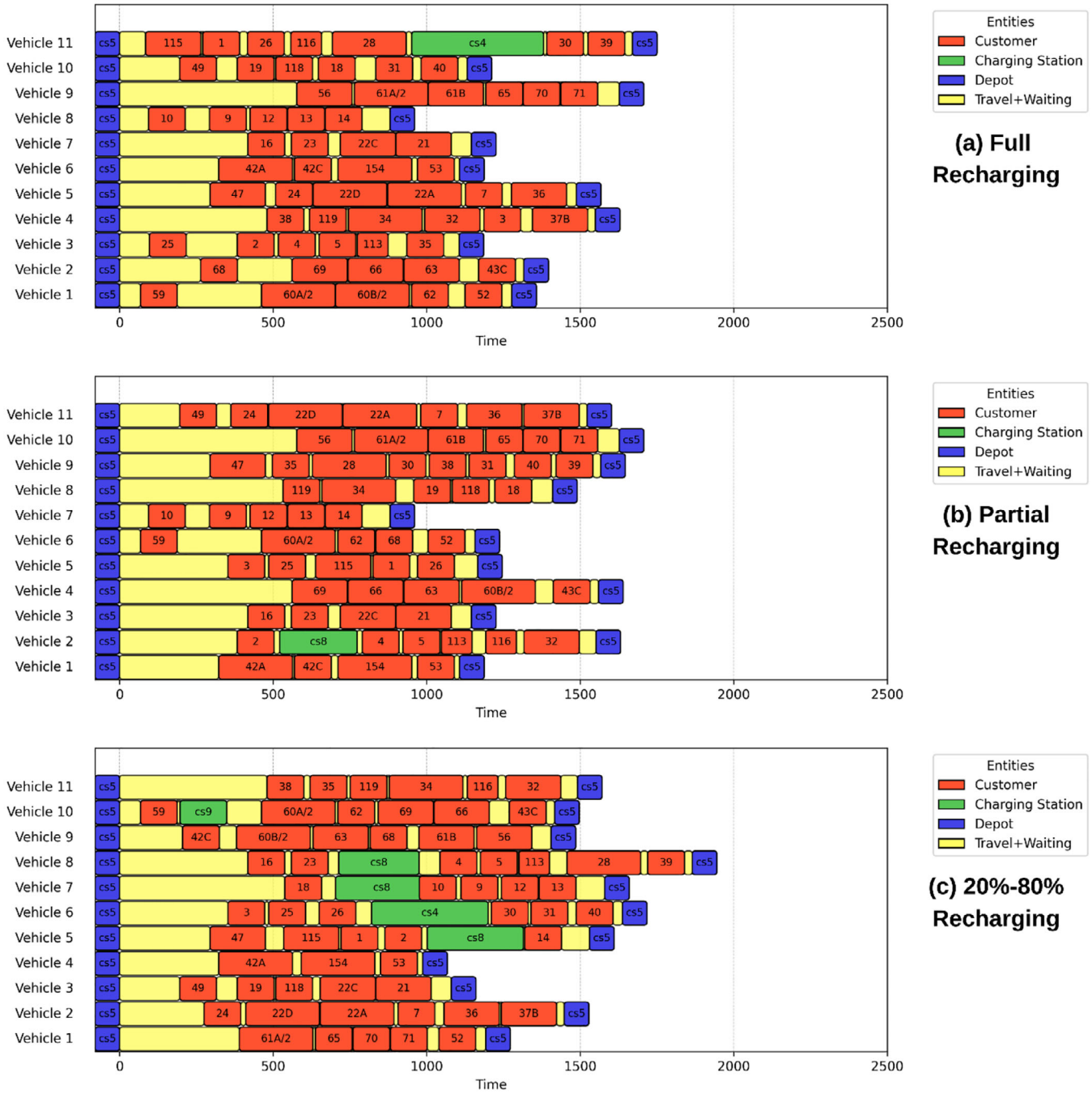

FIGURE 4 | Gantt charts of solutions for minimising total distance.

strategies. A Gantt chart is used to visualise the number of stops at charging stations and the recharging times at each station. Gantt charts of solutions for minimising total distance are given in Figure 4.

When the problem is solved to minimise the total distance for the problem with 60 customers, the customers' demands are met with 11 routes. When using the full and partial recharging strategy, one recharging was required during the routes. When the 20–80 partial charge strategy is used, a total of 5 recharges are required. Gantt charts of solutions for minimising total time for the C60 problem set are given in Figure 5.

Figure 5 shows the routes obtained in order to minimise total travel time. When using the full and partial charging strategy, one recharge and two recharges are required during the routes, respectively. In the 20–80 partial charging strategy, a total of 4 recharges are required. Gantt charts of solutions for minimising total energy are given in Figure 6.

Figure 6 shows the routes created under three different charging strategies to minimise energy. In Figure 6, when using the full and partial charging strategy, two recharges are required during the routes, respectively. In the 20–80 partial charge strategy, a total of 5 recharges are required. Total Recharging Times (TRT) and the

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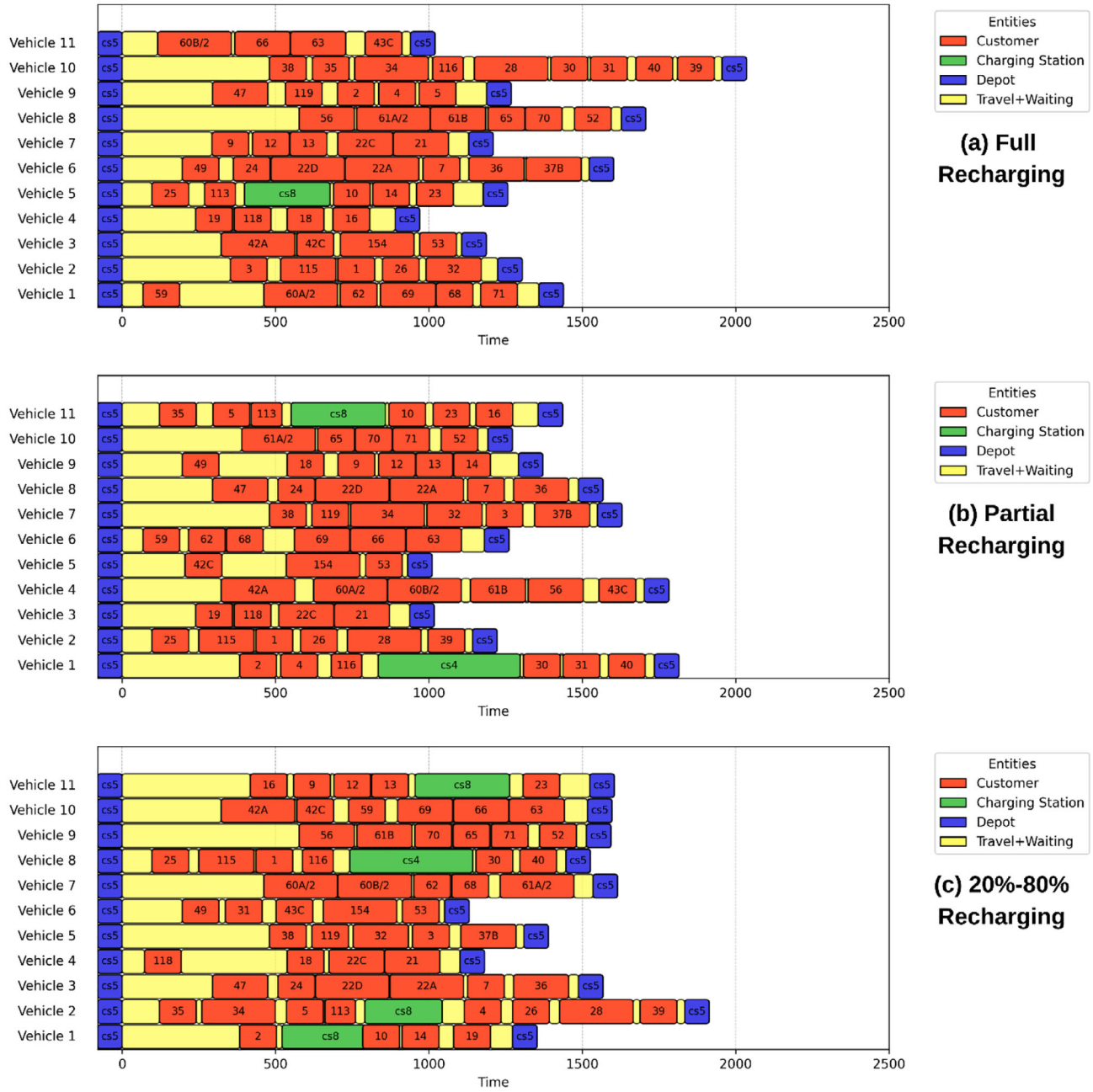

FIGURE 5 | Gantt charts of solutions for minimising total time.

number of Recharges (#Recharge) for all datasets according to the objective functions are given in Table 12.

When the results given in Table 12 are examined, it is understood that total recharging times have the lowest values when the objective function is the total travel time. When the total energy consumption minimization objective is tried to minimize the total energy consumption, the total recharging times are the highest. As a result, the charging strategy to be used in logistics operations should be selected according to the determined objective function. These comparisons allow us to understand which charging strategy more suitable for different objective functions. In logistics operations, the most appropriate strategies can be selected according to the company's goals, and benefits can be achieved in terms of cost and operational efficiency through charging strategy optimization. The analyses obtained within the scope of this study provide an advantage for logistics companies in determining the most appropriate strategy by taking into account the trade-offs of strategies to ensure the optimal balance between customer satisfaction and operational costs while optimizing routes.

5 | Conclusion and Future Works

This study aims to analyse the effectiveness of different charging strategies on various objective functions in route planning for electric vehicles used in last-mile delivery operations. Solutions were generated for three charging strategies (full charge, partial charge, and partial charge in the 20–80% range) in line with

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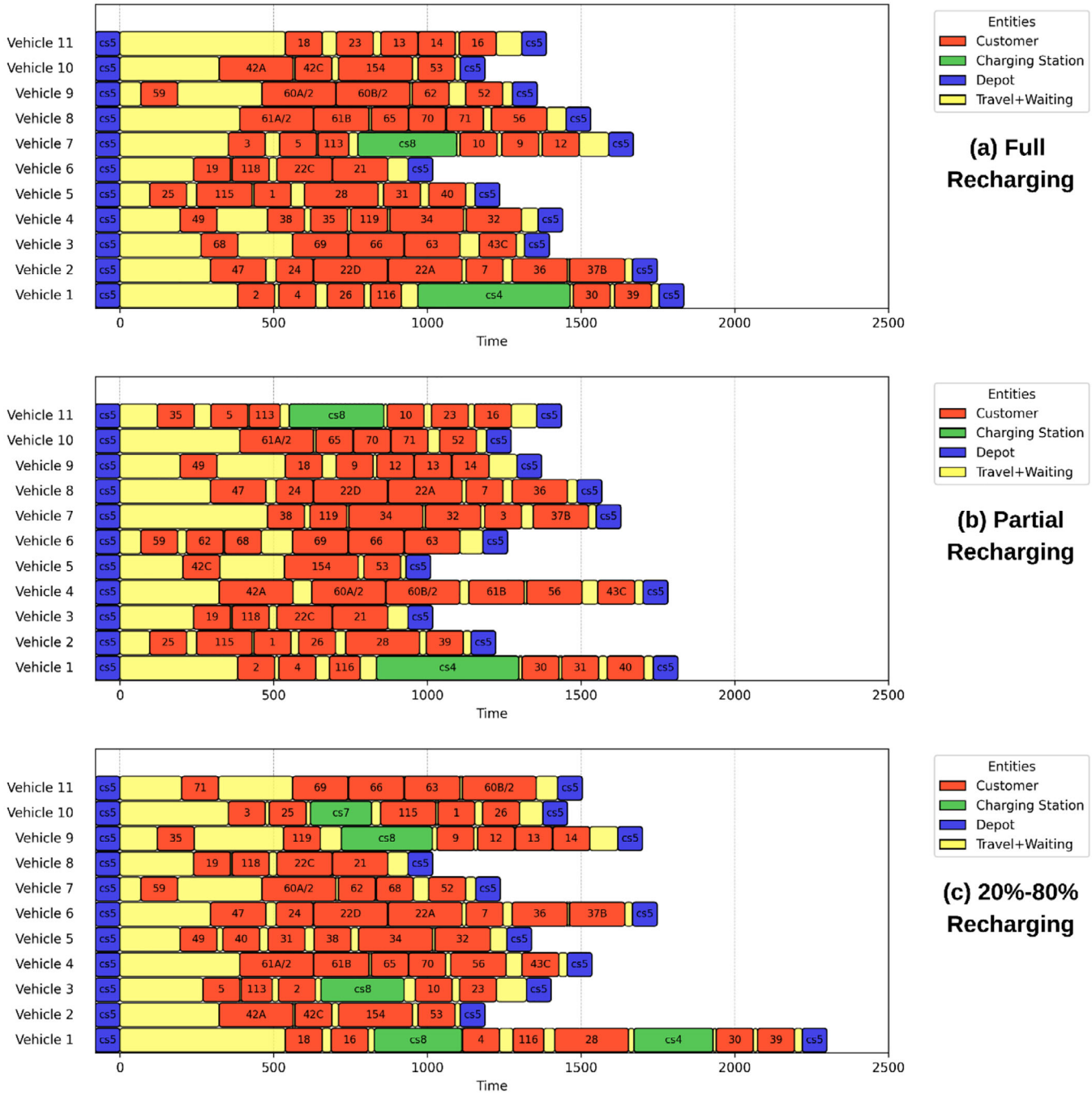

FIGURE 6 | Gantt charts of solutions for minimising total energy.

TABLE 12 | Total recharging times and the number of recharging for all dataset.

	FC		PC		PC_20-80	
	TRT	**# Recharge**	**TRT**	**# Recharge**	**TRT**	**#Recharge**
Min. total distance	5136.87	15	4628.90	15	7935.12	38
Min. total time	2076.39	10	1835.99	8	7893.68	40
Min. total energy	7906.97	23	5488.39	19	9055.19	40
Total	15120.23	48	11953.29	42	24884.00	118

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the objectives of minimising total distance, total time, and total energy using the proposed ALNS algorithm. The results show that each charging strategy offers distinct advantages for different objective functions. The FC strategy requires fewer charging stops and provides an advantage in the total distance travelled through longer recharging times. The PC strategy enables time and energy savings by performing faster charging with shorter recharging times, leading to effective results in minimising time. The 20–80% partial charging strategy requires frequent and long recharging times but stands out for preserving battery life and enhancing energy efficiency. This strategy is especially preferred in energy minimisation but may require trade-offs in total time and distance. For routes aiming to minimise energy consumption, the 20–80% partial charging strategy, which is beneficial for battery health, requires accepting an increase in the makespan for distribution tasks due to the increase in the number of charging. However, the partial charging strategy increases operational efficiency by reducing total travel time on routes where charging needs are low.

The findings obtained within the scope of the study provide practical implications for managers and decision-makers in the logistics sector. Analysing the effects of different charging strategies on total travel distance, total travel time and total energy consumption can help to determine the most appropriate strategies for specific scenarios. In addition, the fact that the proposed ALNS algorithm provides fast and effective solutions to large-scale problems proves that it can be used as a powerful tool to increase operational efficiency and reduce costs in the logistics sector. The findings obtained in this study can provide important guidance to decision-makers on issues such as fleet management, charging infrastructure planning and energy efficiency enhancement. These findings will guide logistics companies in considering charging strategies when making strategic decisions to reduce operational costs and ensure customer satisfaction. In the future, we aim to develop a fleet management system that integrates real traffic data and driver behaviour into electric vehicle route planning to reveal the impact of different charging strategies on various objective functions.

Author Contributions

Sinem Bozkurt Keser: conceptualisation, methodology, formal analysis, writing – original draft, writing – review and editing. **İnci Sarıçiçek:** conceptualisation, methodology, writing – review and editing. **Ahmet Yazıcı:** conceptualisation, writing – review and editing, resources, supervision, project administration and funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available in Mendeley at DOI: 10.17632/7vjzvxh72d.1, reference number [36]. These data were derived from the following resources available in the public domain: https://data.mendeley.com/datasets/7vjzvxh72d/1.

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